

Question

Write a Java program to demonstrate a class **C1** extending another class **C2** and implementing an interface **I1**. Use constructors, method overriding, and method overloading to calculate the salary, bonus, and total salary of an employee.

Steps

1. Create a class **Employee** with a variable **salary**.
2. Create a constructor in **Employee** to initialize the salary.
3. Create a **calculate()** method to display the salary.
4. Overload the **calculate()** method by passing salary as an argument.
5. Create an interface **Bonus** with the method **calculateBonus()**.
6. Create a class **Manager** that **extends Employee** and **implements Bonus**.
7. Create a constructor in **Manager** and call the parent constructor using **super()**.
8. Override the **calculate()** method in **Manager**.
9. Implement the **calculateBonus()** method to calculate 10% bonus.
10. Create another overloaded **calculate()** method that accepts salary and bonus and calculates total salary.
11. Create a **Manager** object in the **main()** method.
12. Display the salary, calculate the bonus, and display the total salary.

Algorithm

1. **Start**
2. Define the `Employee` class with `salary`.
3. Initialize salary using the `Employee` constructor.
4. Define `calculate()` to display salary.
5. Overload `calculate(int salary)`.
6. Define the `Bonus` interface with `calculateBonus(int salary)`.
7. Define `Manager` using:

class Manager extends Employee implements Bonus

8. Initialize the manager's salary using the `Manager` constructor and `super()`.
9. Override `calculate()` to display salary.
10. Implement `calculateBonus()` and calculate:
Bonus = Salary × 10 / 100
11. Overload `calculate(int salary, int bonus)` and calculate:
Total Salary = Salary + Bonus
12. Create a `Manager` object with salary `50000`.
13. Display the salary.
14. Calculate and display the bonus and total salary.
15. **Stop.**

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment2> c::; cd 'c:\
25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessage
e311af8ea878763765323f0f\redhat.java\jdt_ws\Assignment2_4e0b2
Simple Interest: 500
Simple Interest: 1200
Simple Interest: 75
PS C:\Users\venik\OneDrive\Assignmets\Assignment2>
```